



NATIONAL UNIVERSITY OF SINGAPORE

EE4415: Integrated Digital Design

Part II - HW02

A report submitted to

Professor Dr. Xuyano Fong

College of Design and Engineering

EE4915 (Part II)

HW02

(I) Design of static CMOS logic gate

q1) A static CMOS logic gate implements the function

$$Z = \overline{(A+B+C)D+EF}$$

a) Draw the pull-down network for the static CMOS logic gate.

(i) Given function: $Z = \overline{(A+B+C)D+EF}$

Let

$$X = (A+B+C)D+EF$$

Hence,

$$Z = \overline{X}$$

(ii) State PDN rule: The NMOS pull-down network implements X , because the output becomes low when the PDN conducts. In static CMOS.

- OR maps to parallel NMOS
- AND maps to series NMOS

(iii) Break expression into branches:

$$X = (A+B+C)D + EF$$

- $(A+B+C)D$: A, B, C in parallel, then in series with D
- EF : E in series with F

(iv) Final PDN statement:

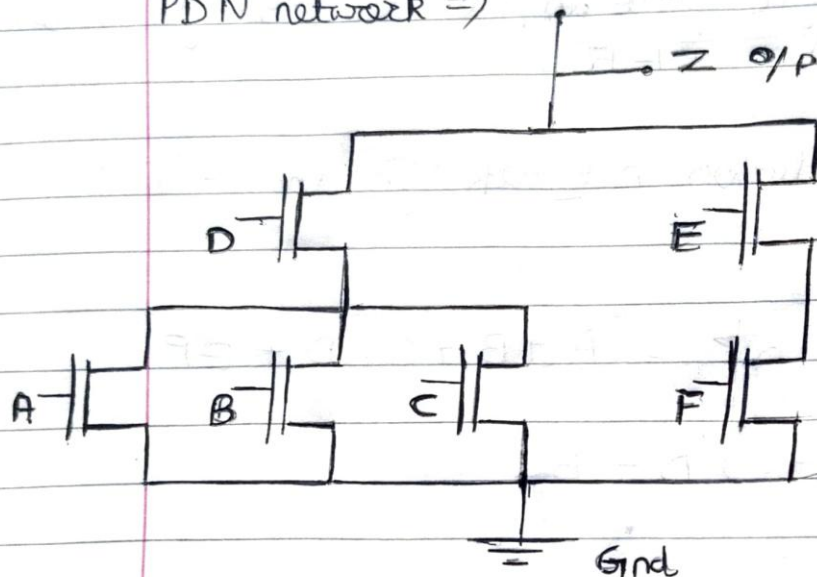


Therefore, the pull-down network is:

$$[D \text{ series } (A \parallel B \parallel C)] \parallel [E \text{ series } F]$$

(v) Draw the circuit:

PDN network $\Rightarrow$



b) Draw the pull-up network for the static CMOS logic gate.

(i) CMOS duality rule: In static CMOS, the pull-up and pull-down networks are dual networks:

- series in PDN becomes parallel in PUN
- parallel in PDN becomes series in PUN

(ii) Starting from PDN:

From part (a),

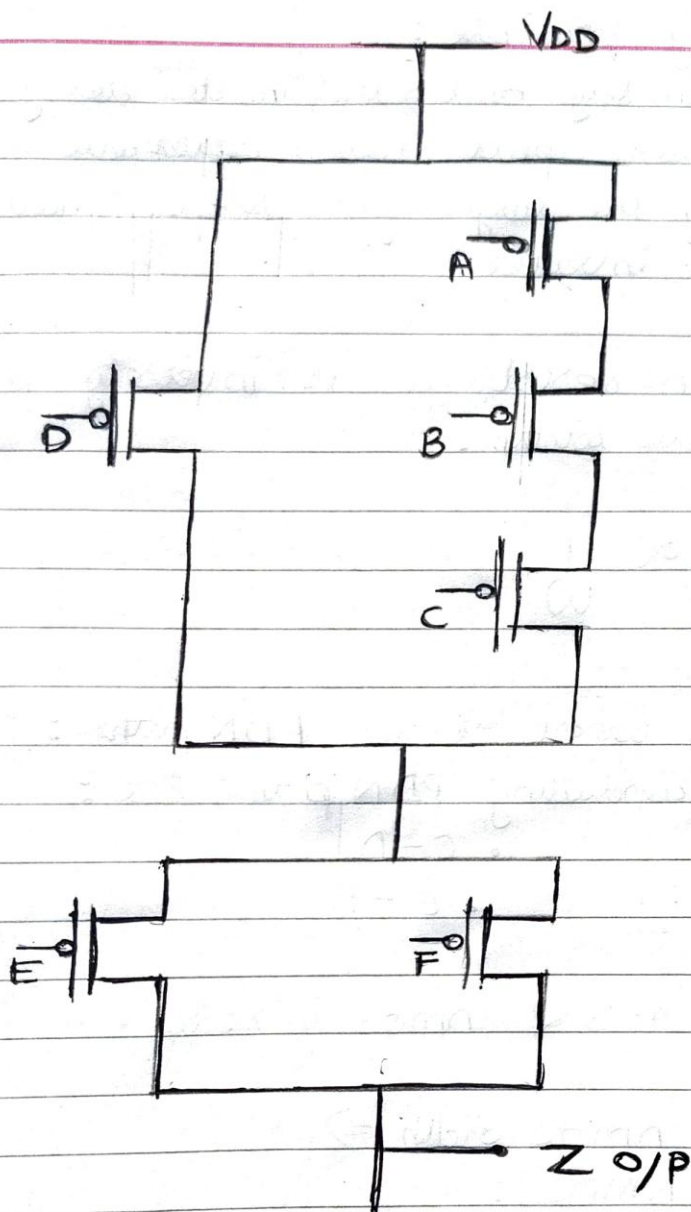
$$PDN = [D \text{ series } (A \parallel B \parallel C)] \parallel [E \text{ series } F]$$

(iii) Take dual:

Therefore, the PUN is,

$$[D \parallel (A \text{ series } B \text{ series } C)] \text{ series } (E \parallel F)$$

(iv) Draw PUN $\Rightarrow$



c) Derive the sizes of all the transistors in the static CMOS logic gate so that the worst-case propagation delays approximately match those of reference static CMOS inverter.

(i) Reference inverter:

$$\left(\frac{W}{L}\right)_N = 1, \quad \left(\frac{W}{L}\right)_P = 3$$

$$R_N = R_P = 2.000 \text{ k}\Omega$$

(ii) Stating sizing principle:

To approximately match the inverter delay, the worst case equivalent pull-down resistance and pull-up resistance of the logic gate should match those of the reference inverter.

Assume on-resistance is inversely proportional to transistor width.

$$R \propto \frac{1}{W}$$

(iii) Identifying worst-case PDN paths:

Possible conducting PDN paths are:

- A-D
- B-D
- C-D
- E-F

Each path has 2 NMOS in series.

(iv) Determining NMOS width $\Rightarrow$

Reference NMOS:

- width = 1
- Resistance = 2.000 k Ω

To keep a 2-stack equal to 2.000 k Ω , each device should have:

$$R_{\text{each}} = 1.000 \text{ k}\Omega$$

Hence, each NMOS should be 2 times the reference width:

$$(W/L)_{nA} = (W/L)_{nB} = (W/L)_{nC} = (W/L)_{nD} =$$

$$(W/L)_{nE} = (W/L)_{nF} = 2$$

(v) Resistance check:

Each NMOS resistance:

$$\frac{2000}{2} = 1000 \text{ k}\Omega$$

Total for 2 series devices:

$$1000 + 1000 = 2000 \text{ k}\Omega$$

That completes NMOS sizing.

(vi) For PMOS:

PUN expression $\Rightarrow [D \parallel (A \text{ series } B \text{ series } C)] \text{ series } (E \parallel F)$

(vii) Identifying worst-case pull-up path:

Possible pull-up paths include:

- D-E
- A-B-C-E
- D-F
- A-B-C-F

Worst-case path is:

$$A-B-C-E \quad \text{or} \quad A-B-C-F$$

which has 4 PMOS in series.

(viii) Deriving PMOS sizes for worst case:

Reference PMOS:

- width = 3
- resistance = 2000 k Ω

For 4 series PMOS to give total 2000 k Ω :

$$R_{\text{each}} = 0.500 \text{ k}\Omega$$

So, each PMOS in the worst path is 4 times the reference PMOS width:

$$4 \times 3 = 12$$

Thus,

$$(W/L)_{PA} = (W/L)_{PB} = (W/L)_{PC} = (W/L)_{PE} = (W/L)_{PF} = 12$$



(ix) Sizing PMOS D :

The path through D with E or F has only 2 PMOS in series.

So, D only needs to be 2 times the reference PMOS width:

$$(W/L)_{PD} = 6$$

(x) Final PMOS answer :

$$(W/L)_{PA} = (W/L)_{PB} = (W/L)_{PC} = 12$$

$$(W/L)_{PD} = 6$$

$$(W/L)_{PE} = (W/L)_{PF} = 12$$

(xi) Summary table :

Transistor	Type	Size (W/L)
A	NMOS	2
B	NMOS	2
C	NMOS	2
D	NMOS	2
E	NMOS	2
F	NMOS	2
A	PMOS	12
B	PMOS	12
C	PMOS	12
D	PMOS	6
E	PMOS	12
F	PMOS	12



d) Draw logic graphs for the pull-up and pull-down networks and clearly show whether a consistent Euler path can be found.

(i) PDN and PUN expressions:

$$\text{PDN} = [(A \parallel B \parallel C) \text{ series } D] \parallel [E \text{ series } F]$$

$$\text{PUN} = [D \parallel (A \text{ series } B \text{ series } C)] \text{ series } (E \parallel F)$$

(ii) Drawing PDN graph:

Rule: • Every electrical node becomes a graph node

• every NMOS transistor becomes a graph edge

PDN nodes = Σ

X: node between D and A/B/C part

Y: node between E & F

Gnd

PDN edges = $D = \Sigma \leftrightarrow X$

A: $X \leftrightarrow \text{Gnd}$

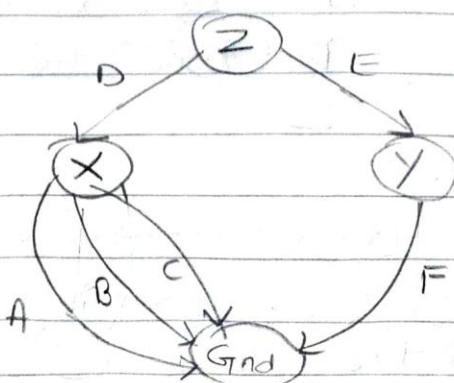
B: $X \leftrightarrow \text{Gnd}$

C: $X \leftrightarrow \text{Gnd}$

E: $\Sigma \leftrightarrow Y$

F: $Y \leftrightarrow \text{Gnd}$

PDN graph $\Rightarrow$



(iii) Checking Euler path condition for PDN:

Calculate degree of each node:

PDN vertex degrees:

$$\deg(Z) = 2$$

$$\deg(X) = 4$$

$$\deg(Y) = 2$$

$$\deg(\text{Gnd}) = 4$$

Since all vertices have even degree, the PDN graph has an Euler path. In fact, it has an Euler circuit.

One valid PDN traversal:

$$Z \xrightarrow{D} X \xrightarrow{A} \text{Gnd} \xrightarrow{B} X \xrightarrow{C} \text{Gnd} \xrightarrow{F} X \xrightarrow{E} Z$$

So, one valid PDN transistor order is,
D, A, B, C, F, E

(iv) Drawing PDN graph:

Rule: • every electrical node becomes a graph node
• every PMOS transistor becomes a graph edge

PDN nodes: VDD

N_1 : node between A & B

N_2 : node between B & C

M: node between two PDN blocks

Z

PDN edges: D: VDD $\leftrightarrow$ M

A: VDD $\leftrightarrow$ N_1

B: $N_1 \leftrightarrow N_2$

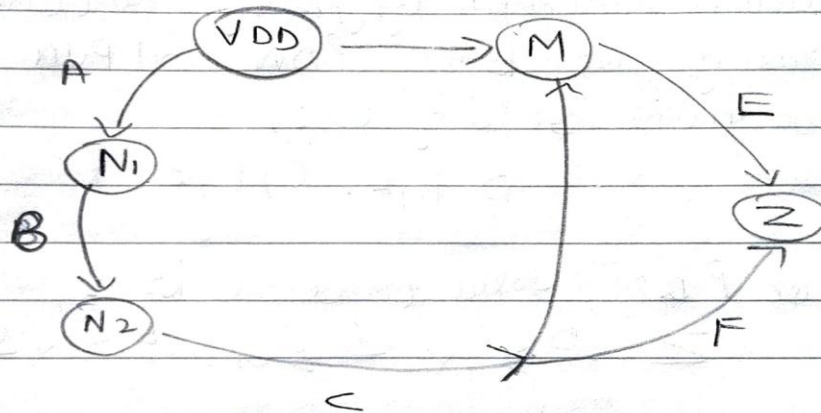


$$C : N_2 \leftrightarrow M$$

$$E : M \leftrightarrow Z$$

$$F : M \leftrightarrow Z$$

PUN graph =



(v) Check Euler path condition for PUN:

Calculate degree of each node:

PUN vertex degrees:

$$\deg(VDD) = 2$$

$$\deg(N_1) = 2$$

$$\deg(N_2) = 2$$

$$\deg(M) = 4$$

$$\deg(Z) = 2$$

vertices

Since all ~~degrees~~ have even degree, the PUN graph also has a Euler path. In fact, it also has a Euler circuit -

One valid PUN traversal:

$$VDD \xrightarrow{A} N_1 \xrightarrow{B} N_2 \xrightarrow{C} M \xrightarrow{E} Z \xrightarrow{F} M \xrightarrow{D} VDD$$

So, one valid PUN transistor order is,
A, B, C, E, F, D

Another valid PUN traversal:

$$M \xrightarrow{D} VDD \xrightarrow{A} N_1 \xrightarrow{B} N_2 \xrightarrow{C} M \xrightarrow{F} Z \xrightarrow{E} M$$

Edge order : D, A, B, C, F, E

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(vi) checking for consistent Euler path:
Comparing 2 valid orders.

Since multiple valid Euler traversals may exist for each graph, the existence of a consistent Euler path must be checked by finding a common consistent ordering for both PUN and PUN. Here such common ordering is:

D, A, B, C, F, E

For PUN, valid traversal is:

$$Z \xrightarrow{D} X \xrightarrow{A} Gnd \xrightarrow{B} X \xrightarrow{C} Gnd \xrightarrow{F} Y \xrightarrow{E} Z$$

For PUN, valid traversal is:

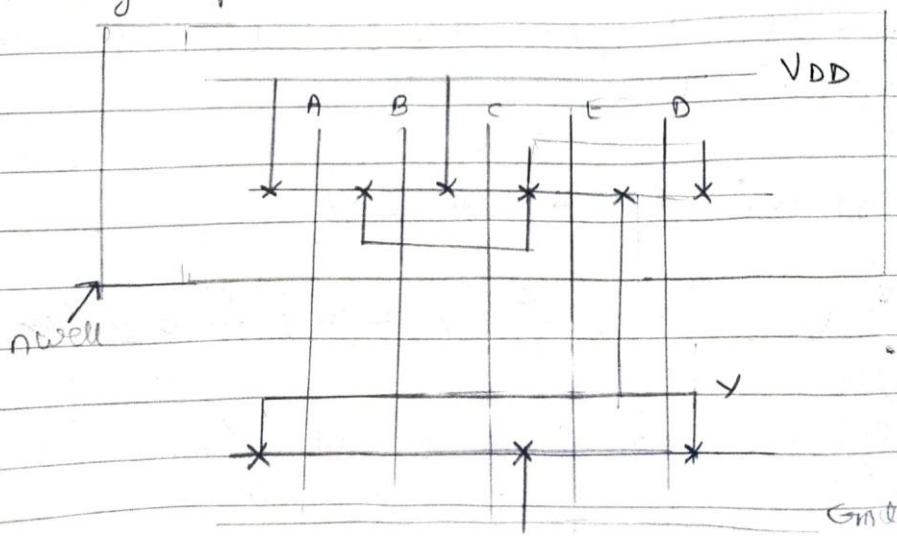
$$M \xrightarrow{D} VDD \xrightarrow{A} N_1 \xrightarrow{B} N_2 \xrightarrow{C} M \xrightarrow{F} Z \xrightarrow{E} M$$

Hence, the pull-up and pull-down networks do have a consistent Euler path with common ordering:

D, A, B, C, F, E

Hence, a fully uninterrupted common diffusion ordering is possible.

Q 2)



Explain what the type (ie static, saturated or dynamic) of the logic gate shown in the figure is by clearly drawing the transistor-level schematic for it.

Ans. The given stick diagram is decoded by identifying the PMOS region, the NMOS region, the poly gate lines, and the metal interconnections. The upper diffusion strip lies inside the n -well, so it forms the PMOS pull-up network (PUN). The lower diffusion strip forms the NMOS pull-down network (PDN). The vertical poly lines labeled A, B, C, E and D define the transistor gates, while the top and bottom metal rails are connected to VDD and Gnd respectively. The node labeled Y is the output.

a) Pull-down network:

Tracing the lower diffusion and metal connection shows that the output node Y is connected to Gnd through two parallel branches.

- one branch contains NMOS transistors A, B and C in series.
- the other branch contains NMOS transistors D and E in series.

Hence, the pull-down network implements:

$$ABC + DE$$

b) Pull-up network:

Tracing the upper diffusion and metal connection shows that:

- PMOS transistor A, B and C are connected in parallel between VDD and an internal node X.



- PMOS transistors D and E are connected in parallel between node X and the output Y.

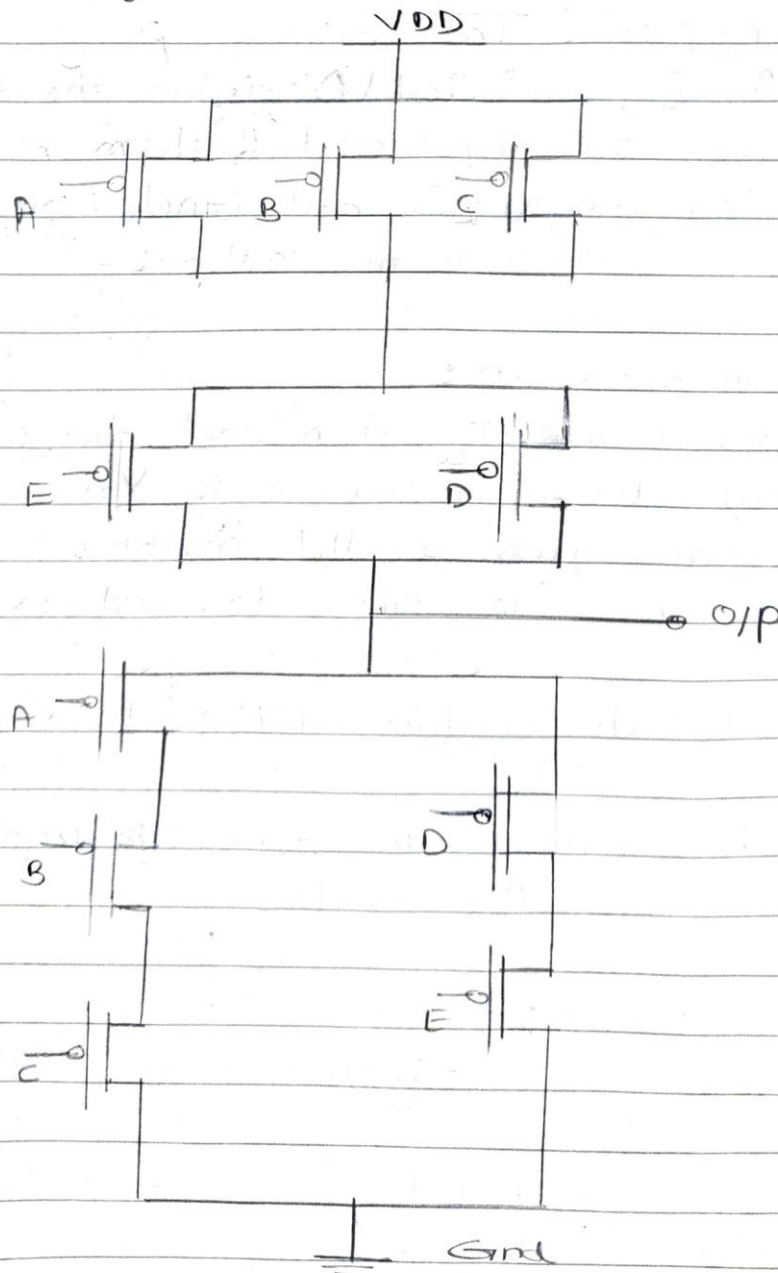
Therefore, the pull-up network is:

$(A \parallel B \parallel C) \text{ series } (D \parallel E)$

This is the dual of the NMOS pull-down network as expected for a static CMOS logic gate.

c) Transistor-level schematic:

The transistor-level schematic corresponding to the stick diagram shown below.





d) Output logic junctions :

From the pull-down network, the o/p is :

$$Y = \overline{ABC + DE}$$

Using De Morgan's theorem,

$$Y = \overline{ABC} \cdot \overline{DE}$$

$$Y = (\overline{A} + \overline{B} + \overline{C})(\overline{D} + \overline{E})$$

This matches the pull up network structure obtained from the stick diagram.

e) Type of logic gate :

The logic gate is a :

Static CMOS logic gate

f) Justification :

This gate is classified as static CMOS because:

(i) it contains both a PMOS pull-up network and a NMOS pull-down network.

(ii) the PMOS and NMOS networks are dual networks.

(iii) the output is connected to either VDD or Gnd through a conducting network in steady state.

(iv) there is no clocked precharge/evaluate transistor, so it is not a dynamic gate.

(v) there is no always-on load transistor, so it is not a static gate.



Conclusion: The stick diagram corresponds to a static CMOS gate implementing:

$$Y = \overline{ABC + DE}$$

(II) ~~Dynamic~~ Design of dynamic static CMOS logic gate:

Q3) Consider the Boolean function:

$$Z = \overline{A + B + C + D + E + F + G + H}$$

a) Draw the dynamic CMOS gate that implements Z using a pull-down network.

Step 1: Write the given function $\Rightarrow$

$$Z = \overline{A + B + C + D + E + F + G + H}$$

Step 2: Identify the PDN function $\Rightarrow$

For dynamic CMOS using a pull-down network, the NMOS PDN implements the non-inverted inside function:

$$A + B + C + D + E + F + G + H$$

Step 3: Map the Boolean function to NMOS network $\Rightarrow$
Since for NMOS:

- OR $\rightarrow$ parallel
- AND $\rightarrow$ series

the PDN is 8 NMOS in parallel, with gates driven by A to H.

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Step 4: Adding the clocking clocked transistors $\Rightarrow$

A dynamic gate needs:

- one PMOS precharge transistor from VDD to output node
- one NMOS evaluation transistor from PDN to Gnd.

Use standard clocking:

- precharge when $\text{clk} = 0$
- evaluate when $\text{clk} = 1$

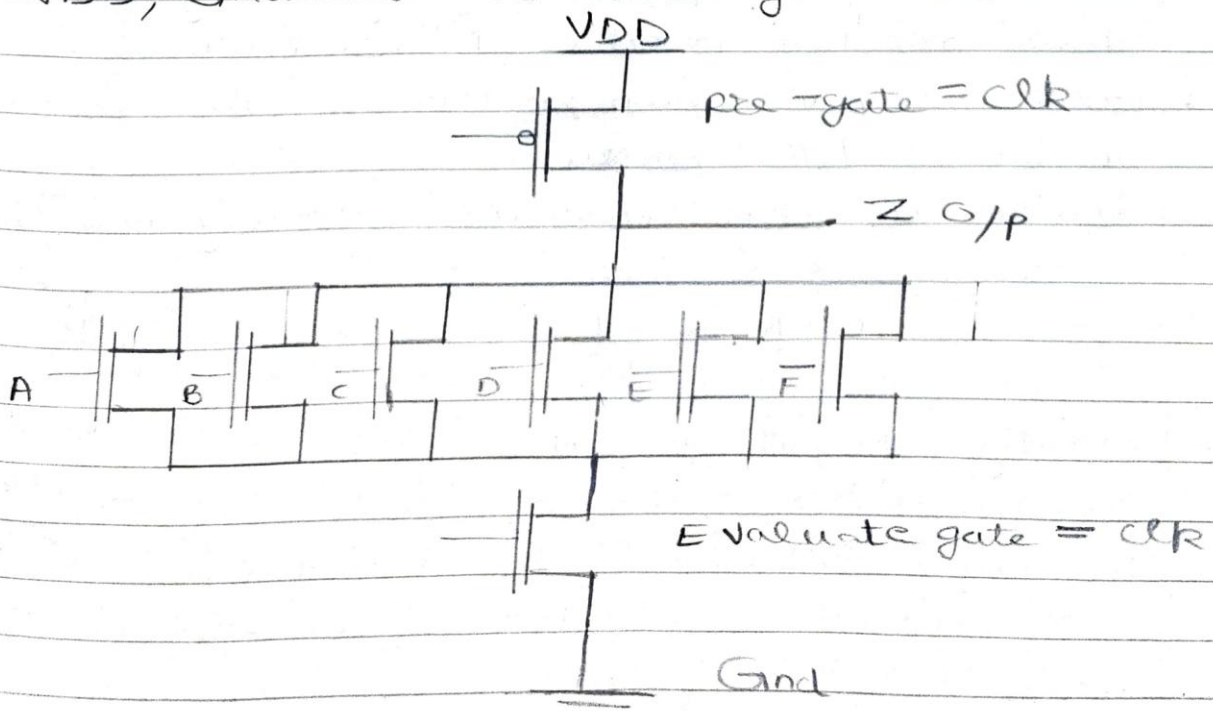
So,

- PMOS precharge gate = clk
- NMOS evaluate gate = clk

Step 5: Draw the final circuit $\Rightarrow$

Draw:

- top PMOS precharge transistor
- output node Z
- 8 parallel NMOS input transistors
- one bottom evaluation NMOS
- VDD, Gnd and clk clearly labeled





Hence, the circuit is a dynamic 8-input NOR gate, since the output is precharged high and discharges during evaluation when any one of A to H is high.

- b) Derive the sizes of all the transistors in the dynamic CMOS logic gate so that the worst-case pull-up and pull-down resistances match that of the reference static CMOS inverter.

Step 1: Write the reference inverter data

$$(W/L)_n = 1, (W/L)_p = 3$$

$$R_n = R_p = 2.000 \text{ K}\Omega$$

Step 2: State the sizing principle

To approximately match the reference inverter delay:

- worst case pull-up resistance of the dynamic gate should match the inverter PMOS resistance.
- worst case pull-down resistance should match the inverter NMOS resistance.

Assume transistor resistance inversely proportional to width:

$$R \propto \frac{1}{W}$$

Step 3: Identify the pull-up path

During precharge, charging occurs only through the single PMOS precharge transistor.

So the worst-case pull-up path contains 1 PMOS transistor.

To match the reference PMOS:

$$(W/L)_{PPse} = 3$$

Step 4: Identify the worst-case pull down path
During evaluation, discharge occurs through:

- core input NMOS
- core evaluation NMOS

So the worst-case pull-down path contains 2 NMOS in series.

Step 5: Derive NMOS sizes

Reference NMOS:

- width = 1
- resistance = 2.000 k Ω

For 2 equal NMOS in series to give total 2.000 k Ω , each should have:

$$R_{each} = 1.000 \text{ k } \Omega$$

Hence, each NMOS should be 2 times wider:

$$(W/L)_{nA} = (W/L)_{nB} = \dots = (W/L)_{nH} = 2$$

$$(W/L)_{nEval} = 2$$

Step 6: Check the resistance

Each width-2 NMOS has:

$$R = \frac{2.000}{2} = 1.000 \text{ k } \Omega$$

So, the discharge per 2 resistances is:

$$1.000 + 1.000 = 2.000 \text{ k } \Omega$$

This matches the reference NMOS resistance.



Step 7: Final Size Summary

$$(w/L)_{pPre} = 3$$

$$(w/L)_{nEval} = 2$$

$$(w/L)_{nA} = (w/L)_{nB} = (w/L)_{nC} = (w/L)_{nD} = 2$$

$$(w/L)_{nE} = (w/L)_{nF} = (w/L)_{nG} = (w/L)_{nH} = 2$$

Summary table $\Rightarrow$

Transistor	Type	Size (w/L)
pPRE	PMOS	3
nEVAL	NMOS	2
nA-nH	NMOS	2

- c) Given the effective leakage current that discharges the output capacitance of the dynamic CMOS logic gate is $1nA$, derive an expression for the largest allowable delay for the evaluation phase.

Step 1: State what limits evaluation time

During evaluation, the PDN should remain OFF, leakage current slowly discharges the dynamic output node.

The maximum evaluation time is the time before the output drops below the minimum valid logic-high level.



Step 2: Use capacitor current relation

$$I = C \frac{\Delta V}{\Delta t}$$

Here,
$$\Delta t = \frac{C \Delta V}{I}$$

Step 3: Define the allowable voltage drop

If the o/p starts at V_{DD} and can fall only to the minimum valid high level V_{IH} , then,

$$\Delta V = V_{DD} - V_{IH}$$

Step 4: Write the full expression

$$t_{\text{eval, max}} = \frac{C_L (V_{DD} - V_{IH})}{I_{\text{leak}}}$$

Given: $I_{\text{leak}} = 1 \text{ nA}$

So,

$$t_{\text{eval, max}} = \frac{C_L (V_{DD} - V_{IH})}{1 \text{ nA}}$$

d) If the largest load capacitance is 50 pF , derive the shortest and longest allowable clock period (and hence, the fastest and slowest clock frequency) that we can operate the dynamic logic gate at

- (i) if the duty cycle of the clock is 50%.
- (ii) if we don't constraint the duty cycle of the clock



Step 1: Write the load capacitance
 $C_L = 50.00 \text{ pF}$

Step 2: Use the matched resistance from part (b)
From part (b):

$$R_{pu} = R_{pd} = 2.000 \text{ k}\Omega$$

Step 3: Calculate the RC time

$$RC = (2.000 \times 10^3)(50.00 \times 10^{-12})$$

$$\boxed{RC = 100.0 \text{ ns}}$$

This is the characteristic time for either precharge or worst-case evaluation.

Step 4: derive shortest clock period

To operate correctly, one cycle must allow:

- one precharge interval
- one evaluation interval

Using the first-order RC estimate:

$$T_{\min} = 100.0 \text{ ns} + 100.0 \text{ ns} \\ = 200.0 \text{ ns}$$

Hence,

$$f_{\max} = \frac{1}{T_{\min}} = \frac{1}{200.0 \text{ ns}} = 5.000 \text{ MHz}$$

$$f_{\max} = 5.000 \text{ MHz}$$

(case i): 50% duty cycle

Step 5: Write the phase relation



If duty cycle is 50% :

$$t_{p, \text{re}} = t_{\text{eval}} = \frac{T}{2}$$

Step 6: Shortest allowable period

Each half-cycle must be at least 100.0 ns :

$$\frac{T_{\min}}{2} = 100.0 \text{ ns}$$

So,

$$\boxed{T_{\min} = 200.0 \text{ ns}}$$

Thus,

$$\boxed{f_{\max} = 5.000 \text{ MHz}}$$

Step 7: Longest allowable period :

Evaluation half-cycle cannot exceed the maximum from part (c) :

$$\frac{T_{\max}}{2} \leq t_{\text{eval}, \max}$$

Hence,

$$\boxed{T_{\max} = 2 t_{\text{eval}, \max}}$$

So,

$$\boxed{f_{\min} = \frac{1}{2 t_{\text{eval}, \max}}}$$

Case (ii) : duty cycle not constrained

Step 8: Shortest allowable period

Without duty-cycle constraint, the minimum precharge and minimum evaluation times are each 100.0 ns , so:

$$T_{\min} = 200.0 \text{ ns}$$

and

$$f_{\max} = 5.000 \text{ MHz}$$

Step 9: Largest allowable period

Now evaluation can be as long as $t_{\text{eval, max}}$ while precharge only needs the minimum 100.0 ns . So,

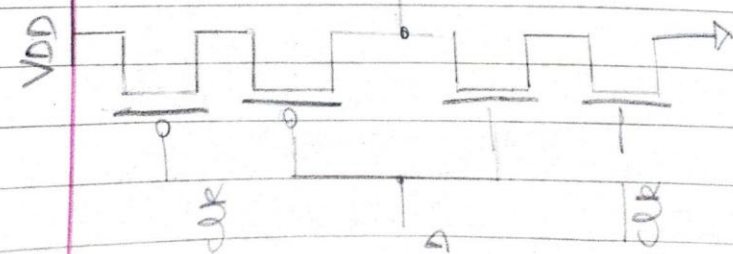
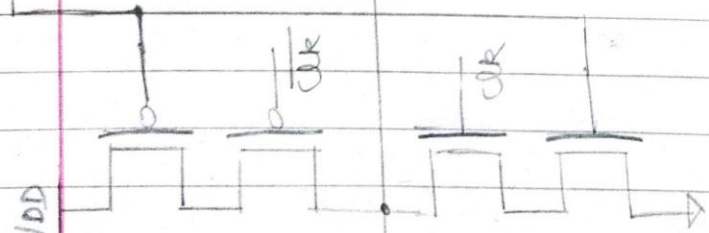
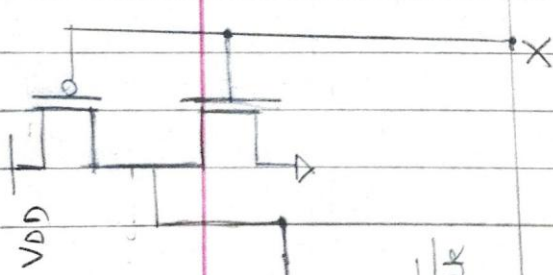
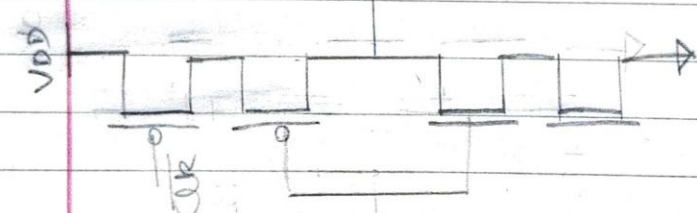
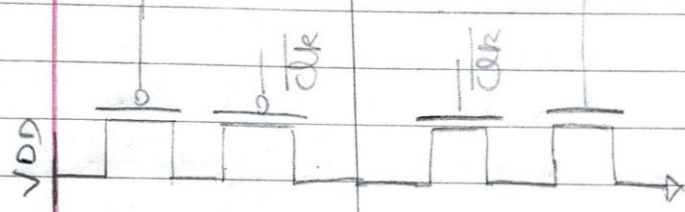
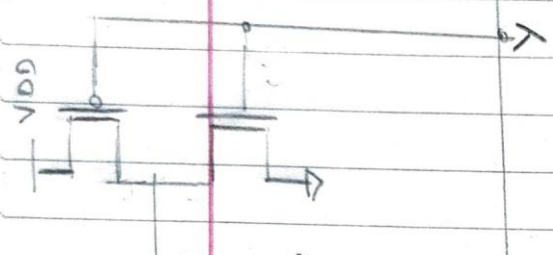
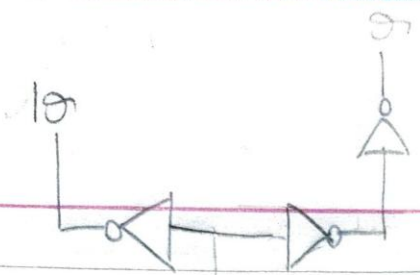
$$T_{\max} = 100.0 \text{ ns} + t_{\text{eval, max}}$$

Thus,

$$f_{\min} = \frac{1}{100.0 \text{ ns} + t_{\text{eval, max}}}$$

(III) Analysis of sequential circuits

Q4) Refer to the figure below and answer the questions that follow.





a)

clearly explain whether the circuit shown in fig 4 is a latch or a flip flop - Use waveforms for the node voltages to aid your explanations if they help.

Ans. Identification of circuit operation:

The circuit in Fig 4 is a flip flop, not a latch. More specifically based on its structure and it's similarity to the lecture example "Unit 5 Sequential circuits, Slide 38", it is a Pseudo-static dynamic positive-edge triggered D flip-flop / register. The circuit contains two clock controlled stages connected in cascade, with internal nodes X and Y, followed by output inverters that generate Q and $\overline{Q}$. The weak feedback inverters connected to the internal nodes indicate a pseudo-static dynamic implementation, which improves robustness compared with a purely dynamic register.

During the low phase of the clock, the master stage is transparent and the input D is sampled onto node X, while the slave stage holds its previous value. When the clock makes a low-to-high transition, the master stage stops sampling and stores the sampled value, while the slave stage becomes transparent and transfers the stored value to node Y. The output is then produced from node Y through the output inverter chain. Therefore, the output



changes only at the rising edge of the clock which confirms that the circuit is a positive-edge triggered flip-flop rather than a level-sensitive latch.

The circuit is a pseudo-static dynamic positive-edge triggered D flip flop

Waveforms :

clk:

D:

X:

(X follows D while master is transparent)

Y:

(Y updates after rising clock edge)

Q:

(Q follows Y after propagation delay)

Waveform explanation :

The input D is independent of the clock and changes at arbitrary times. Node X follows D only during the phase when the master stage



is transparent. Node Y updates only after the rising clock edge when the slave stage becomes transparent. The output Q then follows Y after propagation delay. This confirms positive-edge triggered flip-flop operation.

- b) Analyze the circuit shown in Fig Q4 and derive the delay paths that need to be considered to determine the setup time, hold time and clock-to- Q delay.

Ans. From timing analysis, the important nodes are D , X , Y and Q . The overall signal flow is:

$$D \rightarrow X \rightarrow Y \rightarrow Q$$

The timing path analysis should follow the lecture treatment of dynamic edge-triggered register, since the circuit in Q4 is its pseudo static form. In this structure, the setup time is determined by the master sampling path; the hold time is ideally very small and the clock-to- Q delay is determined by the slave transfer path plus inverter delay.

Setup time:

It is the minimum time before the active clock edge for which the input D must be stable so that the master stage can correctly sample the value and establish it at node X . Therefore, the relevant setup-time path is the

path from input D through the first sampling device and inverter / storage path to node X. It is analogous to:

$$t_{su} \approx t_{d,T} + t_{d,inv}$$

where, $t_{d,T}$ is the delay through the master transfer device and $t_{d,inv}$ is the associated inverter delay.

$$\boxed{\text{Setup path} = D \rightarrow X}$$

Hold time :

It is the minimum time after the clock edge for which the input must remain stable so that the sampled value is not corrupted. For the ideal dynamic edge-triggered register in the lecture, the hold time is approximately zero as the master sampling path turns OFF at the active edge, preventing later input changes from affecting the stored value. Hence, for this circuit, the hold-time path is ~~given~~ again associated with the input path from D to X, but the ideal result is:

$$t_{hold} \approx 0$$

In practice, the non-ideal clock overlap can make the effective hold-time requirement non-zero, but the lecture based ideal



answer is approximately zero.

$$\boxed{\text{Hold path} = D \rightarrow X, t_{\text{hold}} \approx 0}$$

clock-to-Q delay:

It is measured from the active clock edge to the resulting transition at output Q. After the rising edge, the stored value at the master stage is transferred through the slave stage to node Y, and then propagated through the output inverter for Q. Therefore, the relevant clock-to-Q path is:

$$\boxed{\text{Clock-to-Q path} = X \rightarrow Y \rightarrow Q}$$

It is analogous to the lecture expressions,

$$t_{C \rightarrow Q} \approx t_{d, T_2} + t_{d, \text{inv}}$$

where t_{d, T_2} is the delay through the slave transfer device and $t_{d, \text{inv}}$ is the output inverter delay.

Timing summary:

Thus, the timing quantities are determined by:

$$\text{Setup path} = D \rightarrow X, t_{\text{su}} \approx t_{d, T_1} + t_{d, \text{inv}}$$

$$\text{Hold path} = D \rightarrow X, t_{\text{hold}} \approx 0$$

$$\text{Clock-to-Q path} = X \rightarrow Y \rightarrow Q, t_{C \rightarrow Q} \approx t_{d, T_2} + t_{d, \text{inv}}$$



These are the required delay paths for determining the timing behaviour of the sequential circuit.

$D \dashrightarrow [\text{Master stage}] \dashrightarrow X \dashrightarrow [\text{Slave stage}]$

$\phi \dashleftarrow [\text{Output inverter}] \dashleftarrow X \dashleftarrow 1$

Setup path: $D \dashrightarrow X$

Hold path: $D \dashrightarrow X$

Clock-to- ϕ path: $X \dashrightarrow Y \dashrightarrow \phi$

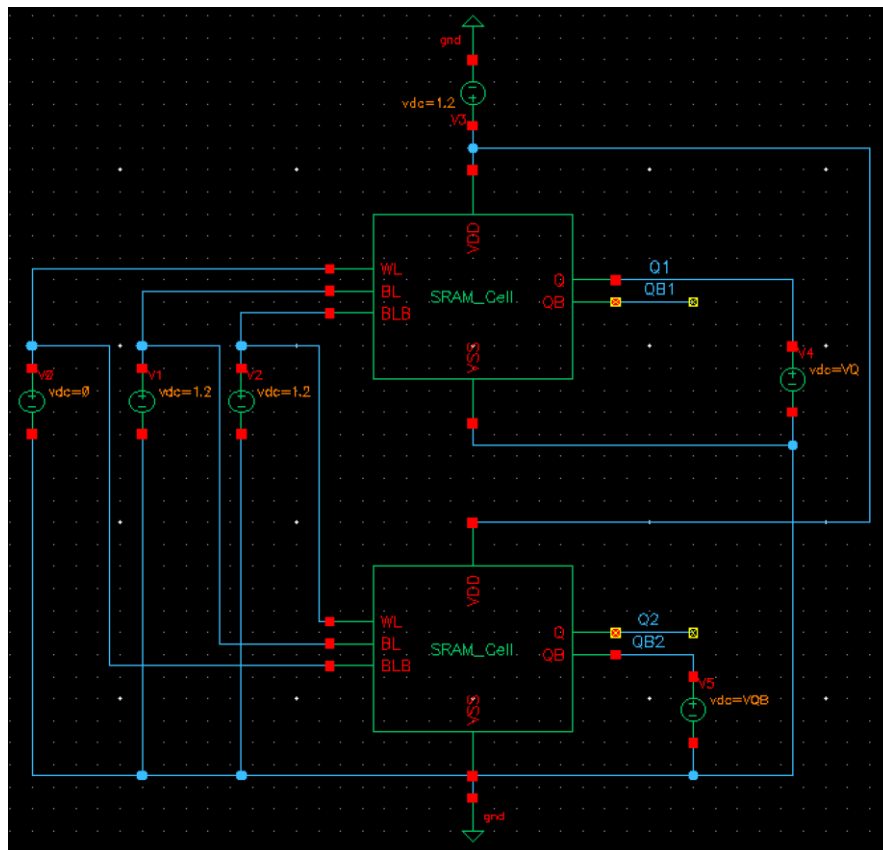
Q5. When designing the 6T static RAM cell, designers want to plot figures of merit for the bitcell to determine its performance. In this question, you will use the FreePDK45 transistor models to investigate the 6T SRAM design issues. The channel length for all MOSFETs in the 6T SRAM is $L = 50$ nm. The W of the pull-up, pull-down and access transistors are 200 nm, 400 nm and 200 nm, respectively. Also, assume that $V_{DD} = 1.2$ V.

[illegible]

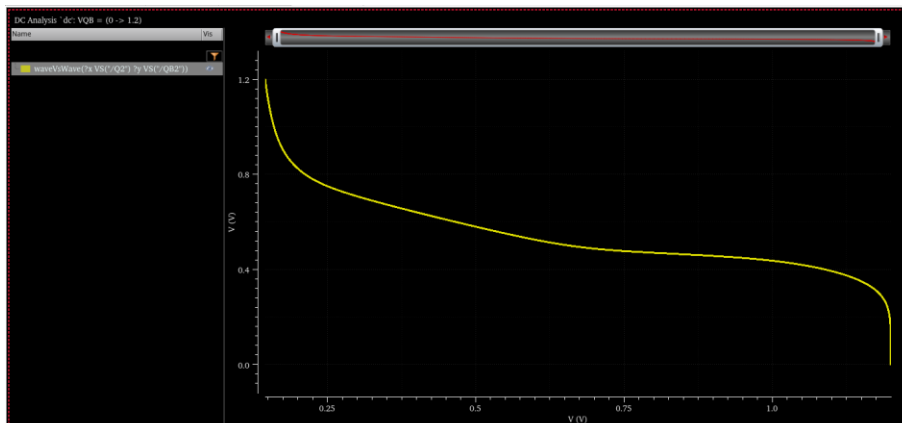
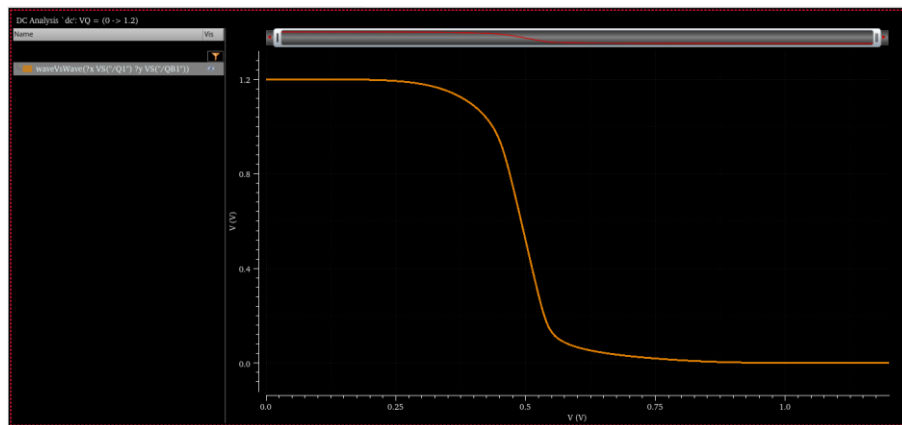
The image displays two side-by-side screenshots of the 'Choosing Analyses' dialog box in ADE Explorer. The left screenshot shows the 'DC Analysis' tab with 'Design Variable' selected as the sweep variable and 'VQ' as the variable name. The right screenshot shows the 'DC Analysis' tab with 'Design Variable' selected as the sweep variable and 'VQB' as the variable name. Both screenshots show the 'Sweep Range' section with 'Start-Stop' selected and a range from 0 to 1.2, and the 'Sweep Type' section with 'Linear' selected and a step size of 0.001.

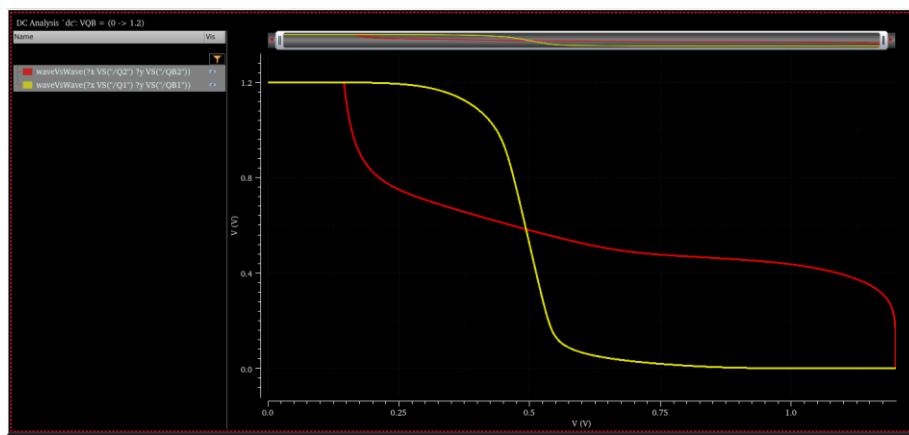
could fit inside each lobe was then estimated graphically using markers placed on the final selected boundary points. Several marker positions were tried graphically, and only the final best-fit points were used for calculation.

TestBench for Hold Operation:



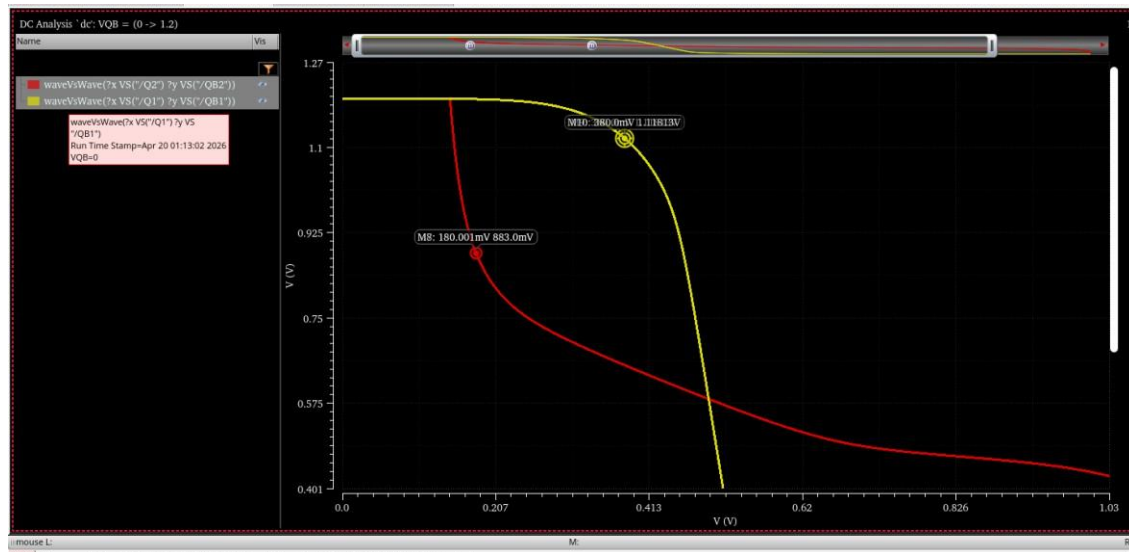
Waveforms:





Final Marker Values Used

For the **left lobe**:



Upper point on yellow curve: (0.3800, 1.1118)

Lower point on red curve: (0.1800, 0.8830)

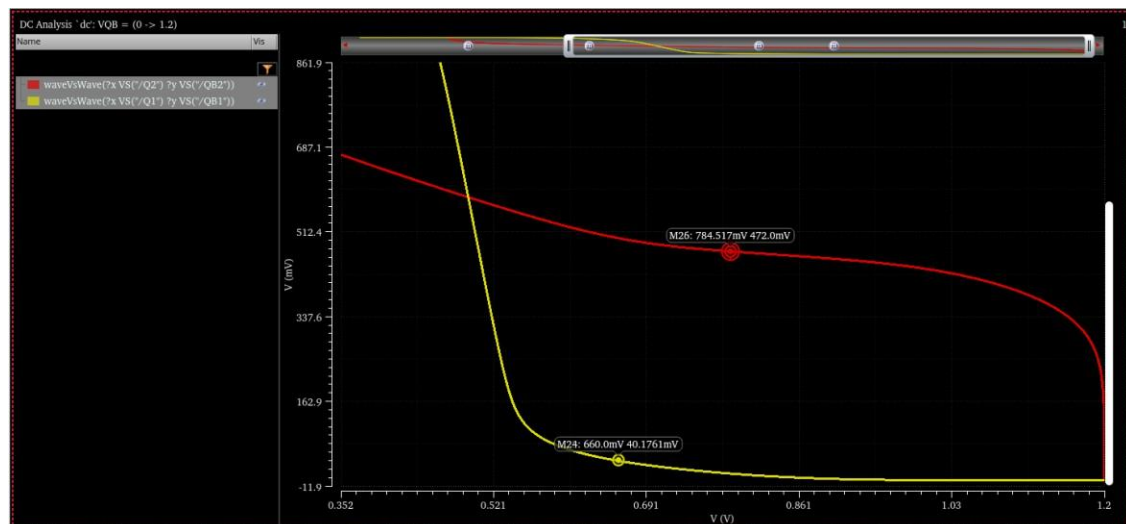
$$\Delta x = 0.3800 - 0.1800 = 0.2000 \text{ V}$$

$$\Delta y = 1.1118 - 0.8830 = 0.2288 \text{ V}$$

Hence, the left-lobe SNM was approximated as:

$$SNM_{left} \approx 0.2000 \text{ V}$$

For the **right lobe**:



Upper point on red curve: (0.7845, 0.4720)

Lower point on yellow curve: (0.6600, 0.0402)

$$\Delta x = 0.7845 - 0.6600 = 0.1245 \text{ V}$$

$$\Delta y = 0.4720 - 0.0402 = 0.4318 \text{ V}$$

Hence, the right-lobe SNM was approximated as:

$$SNM_{right} \approx 0.1245 \text{ V}$$

Result

The hold static noise margin is taken as the smaller of the two lobe values:

$$SNM_{hold} = \min (SNM_{left}, SNM_{right})$$

$$SNM_{hold} = \min (0.2000, 0.1245) = 0.1245 \text{ V}$$

Therefore, the hold static noise margin of the 6T SRAM cell is:

$$SNM_{hold} \approx 0.1245 \text{ V} = 124.5 \text{ mV}$$

Conclusion

The SNM values were estimated graphically from the plotted transfer characteristics using the final selected marker positions. The DC butterfly plot of the 6T SRAM cell under hold condition was obtained successfully. From the final selected marker positions, the left-lobe SNM was estimated to be approximately 0.200 V, while the right-lobe SNM was approximately 0.125 V. Since the hold SNM is defined by the smaller of the two values, the final hold static noise margin of the SRAM cell was found to be **125 mV**.

XX

5(b) Read Static Noise Margin of 6T SRAM Cell

Objective

To perform DC analysis of a 6T SRAM cell under read condition and determine the **read static noise margin (SNM)** from the butterfly plot.

Given Parameters

$$V_{DD} = 1.2 \text{ V}$$

$$V_{WL} = V_{BL} = V_{BLB} = 1.2 \text{ V}$$

$$L_{ch} = 50 \text{ nm}$$

Pull-up transistor width = 200 nm

Pull-down transistor width = 400 nm

Access transistor width = 200 nm

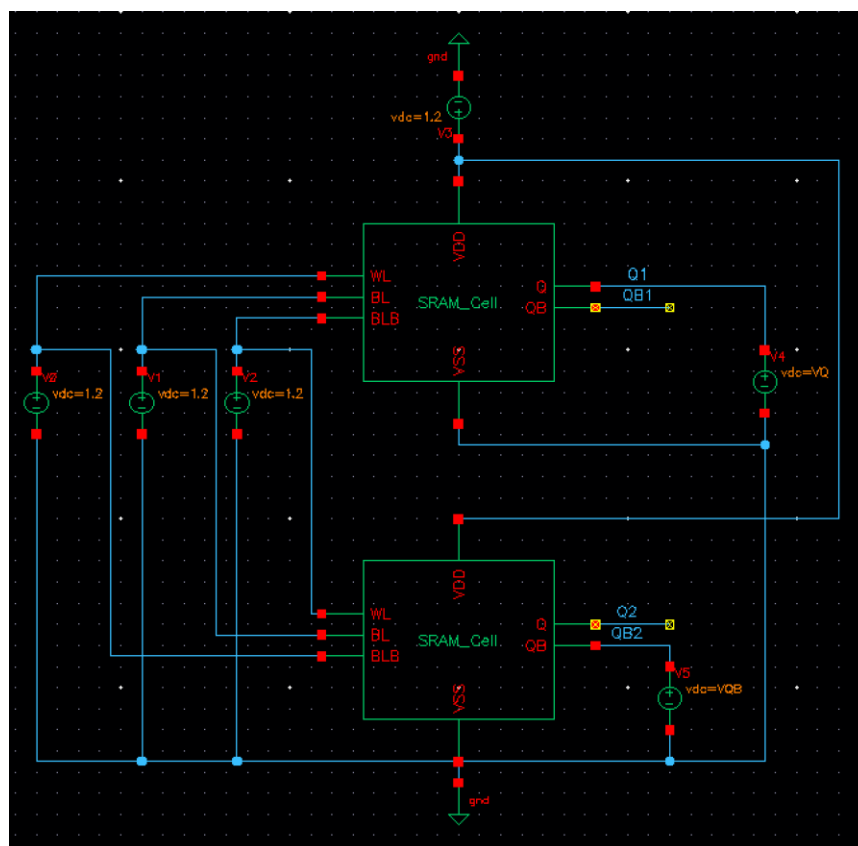
Procedure

A DC analysis was carried out for the 6T SRAM cell under read condition. In this case, the word line and both bit lines were set to 1.2 V. Since the word line was enabled, the access transistors were ON, and the internal storage nodes experienced read disturbance.

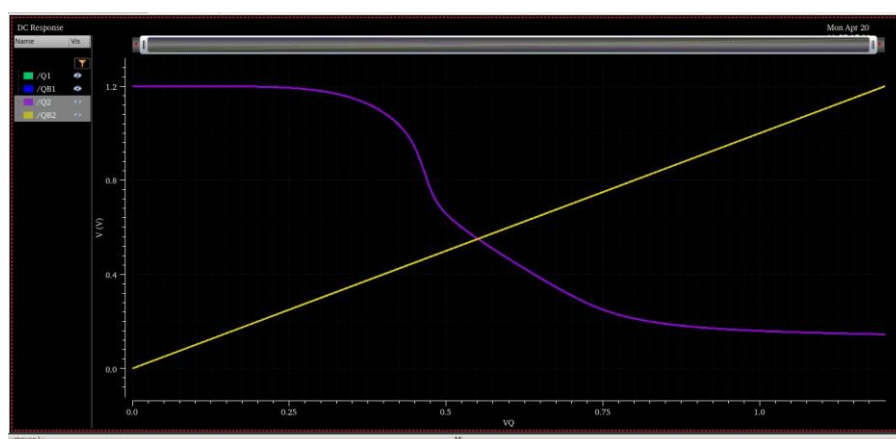
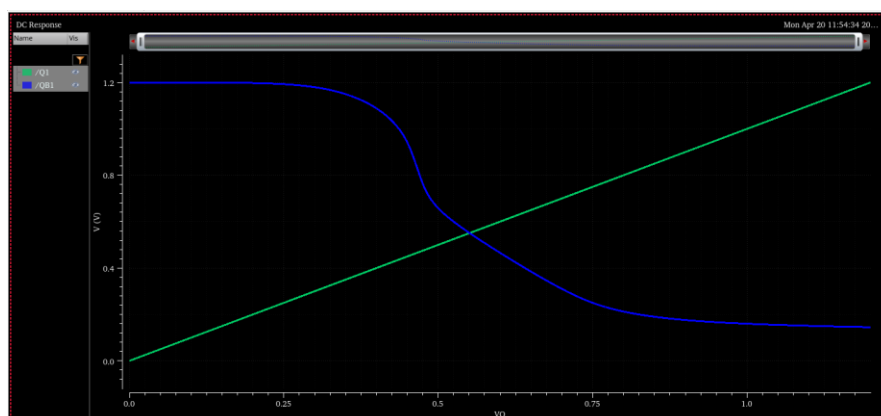
For the first case, the internal storage node voltage V_Q was swept from 0 V to 1.2 V, and the corresponding voltage V_{QB} was measured. For the second case, the complementary node voltage V_{QB} was swept from 0 V to 1.2 V, and the corresponding voltage V_Q was measured.

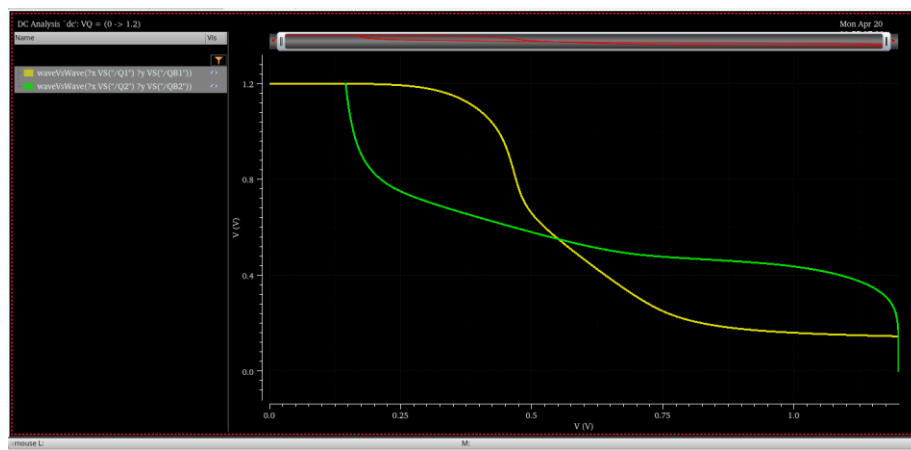
The two voltage transfer characteristics were plotted together on the same axes as V_{QB} versus V_Q to obtain the butterfly plot under read condition. The axes were kept at the same voltage scale to avoid visual distortion. The

TestBench for Read Operation:



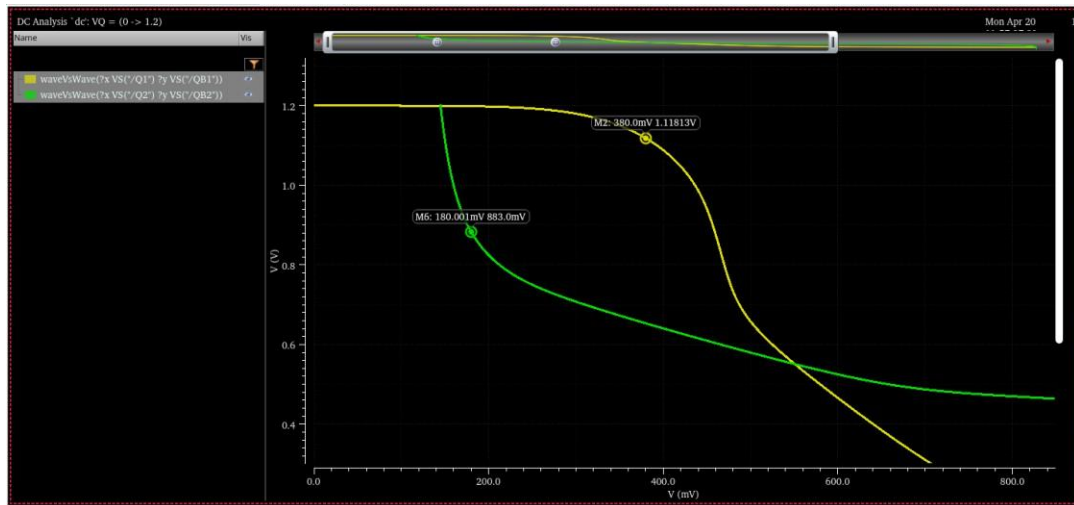
Waveforms:





Final Marker Values Used

For the **left lobe**:



Upper point on yellow curve: (0.3800, 1.1181)

Lower point on green curve: (0.1800, 0.8830)

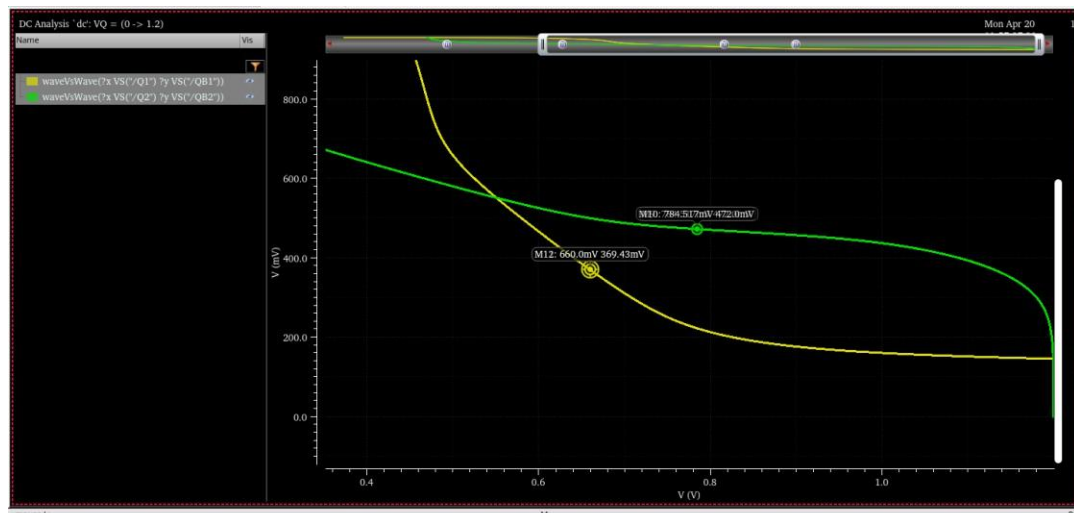
$$\Delta x = 0.3800 - 0.1800 = 0.2000 \text{ V}$$

$$\Delta y = 1.1181 - 0.8830 = 0.2351 \text{ V}$$

Hence, the left-lobe read SNM was approximated as:

$$SNM_{left,read} \approx 0.2000 \text{ V}$$

For the **right lobe**:



Upper point on green curve: (0.7845, 0.4720)

Lower point on yellow curve: (0.6600, 0.3694)

$$\Delta x = 0.7845 - 0.6600 = 0.1245 \text{ V}$$

$$\Delta y = 0.4720 - 0.3694 = 0.1026 \text{ V}$$

Hence, the right-lobe read SNM was approximated as:

$$SNM_{right,read} \approx 0.1026 \text{ V}$$

Result

The read static noise margin is taken as the smaller of the two lobe values:

$$SNM_{read} = \min(SNM_{left,read}, SNM_{right,read})$$

$$SNM_{read} = \min(0.2000, 0.1026) = 0.1026 \text{ V}$$

Therefore, the **read static noise margin of the 6T SRAM cell is:**

$$SNM_{read} \approx 0.1026 \text{ V} = 102.6 \text{ mV}$$

Conclusion

The SNM values were estimated graphically from the plotted transfer characteristics using the final selected marker positions. The DC butterfly plot of the 6T SRAM cell under read condition was obtained successfully. Due to the read disturbance caused by turning ON the access transistors, the butterfly lobes became smaller compared to the hold condition. From the final selected marker positions, the left-lobe read SNM was estimated to be approximately 0.200 V, while the right-lobe read SNM was approximately 0.103 V. Since the read SNM

is defined by the smaller of the two values, the final read static noise margin of the SRAM cell was found to be **103 mV**.

XX

5(c) Write Static Noise Margin of 6T SRAM Cell

Objective

To perform DC analysis of a 6T SRAM cell under write condition and determine the **write static noise margin (SNM)** from the separation between the two plotted curves.

Given Parameters

$$V_{DD} = 1.2 \text{ V}$$

$$V_{WL} = 1.2 \text{ V}$$

$$V_{BL} = 1.2 \text{ V}$$

$$V_{BLB} = 0 \text{ V}$$

$$L_{ch} = 50 \text{ nm}$$

Pull-up transistor width = 200 nm

Pull-down transistor width = 400 nm

Access transistor width = 200 nm

Procedure

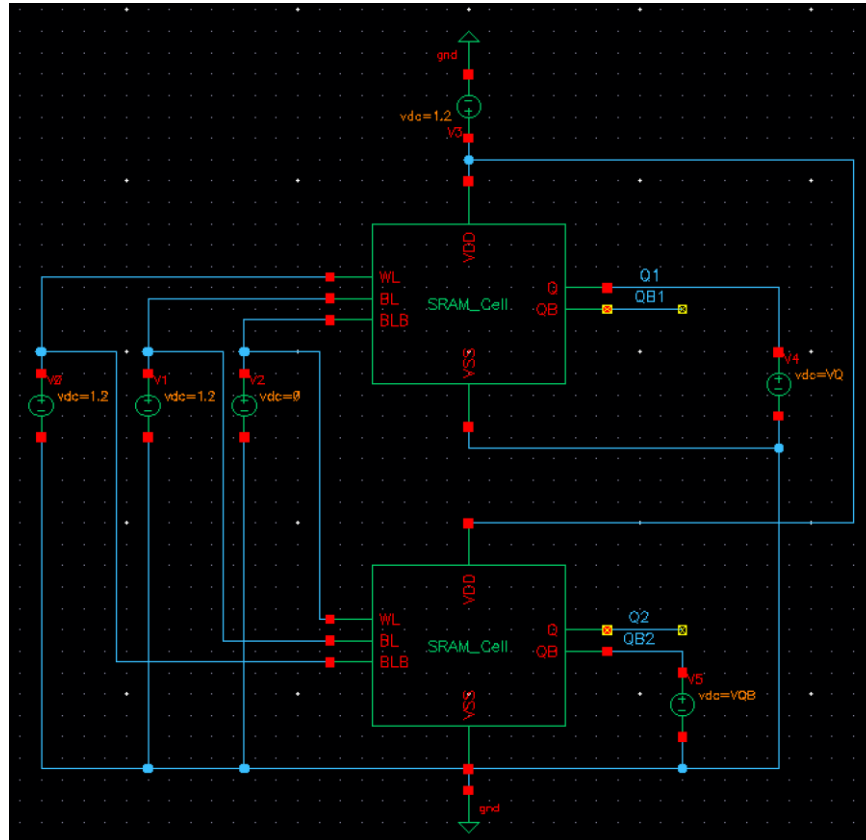
A DC analysis was carried out for the 6T SRAM cell under write condition. In this case, the word line was

enabled, the bit line BL was fixed at 1.2 V, and the complementary bit line BLB was fixed at 0 V. This condition forces the cell toward a write operation and disturbs the internal storage nodes.

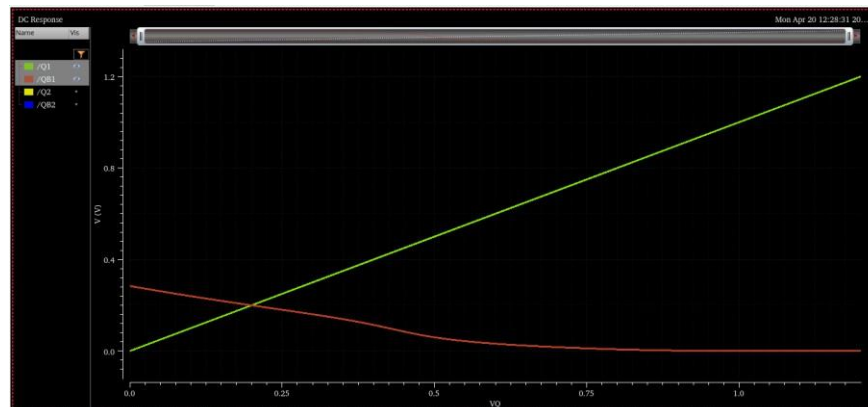
For the first case, the internal storage node voltage V_Q was swept from 0 V to 1.2 V, and the corresponding voltage V_{QB} was measured. For the second case, the complementary node voltage V_{QB} was swept from 0 V to 1.2 V, and the corresponding voltage V_Q was measured.

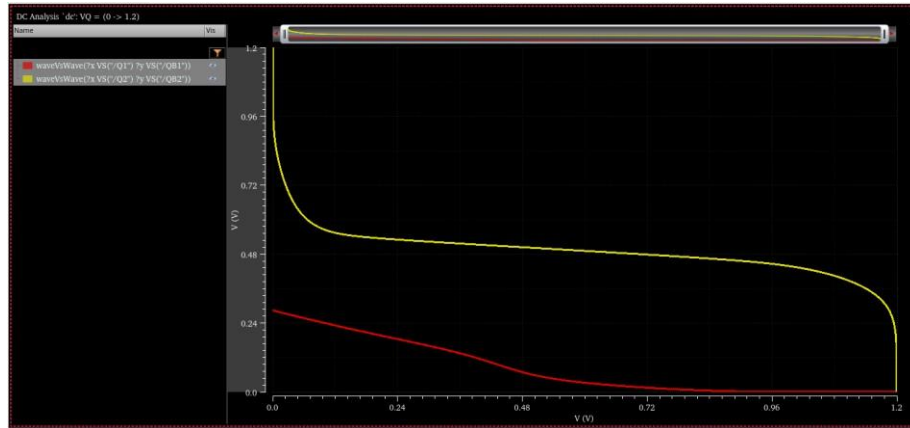
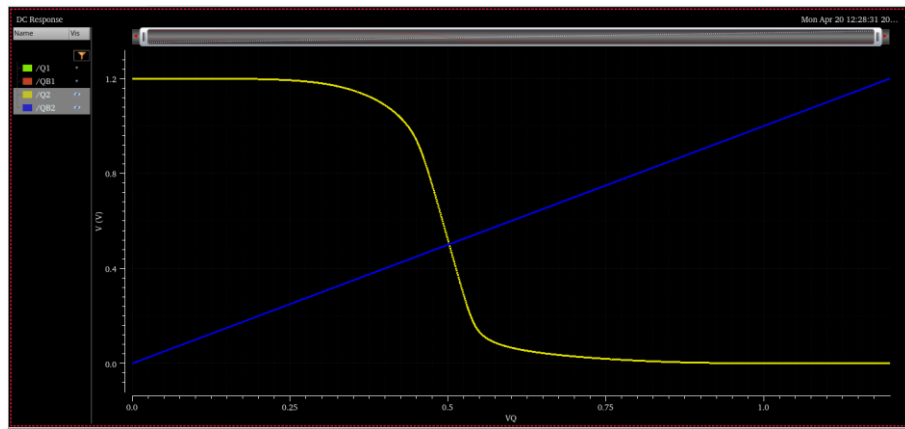
The two voltage transfer characteristics were plotted together on the same graph. Unlike the hold and read conditions, the write condition does not produce the conventional butterfly lobes. Therefore, the write static noise margin was estimated by finding the largest square that could fit in the separation region between the two curves. Several marker positions were tried graphically, and only the final best-fit points were used for calculation.

TestBench for Write Operation:

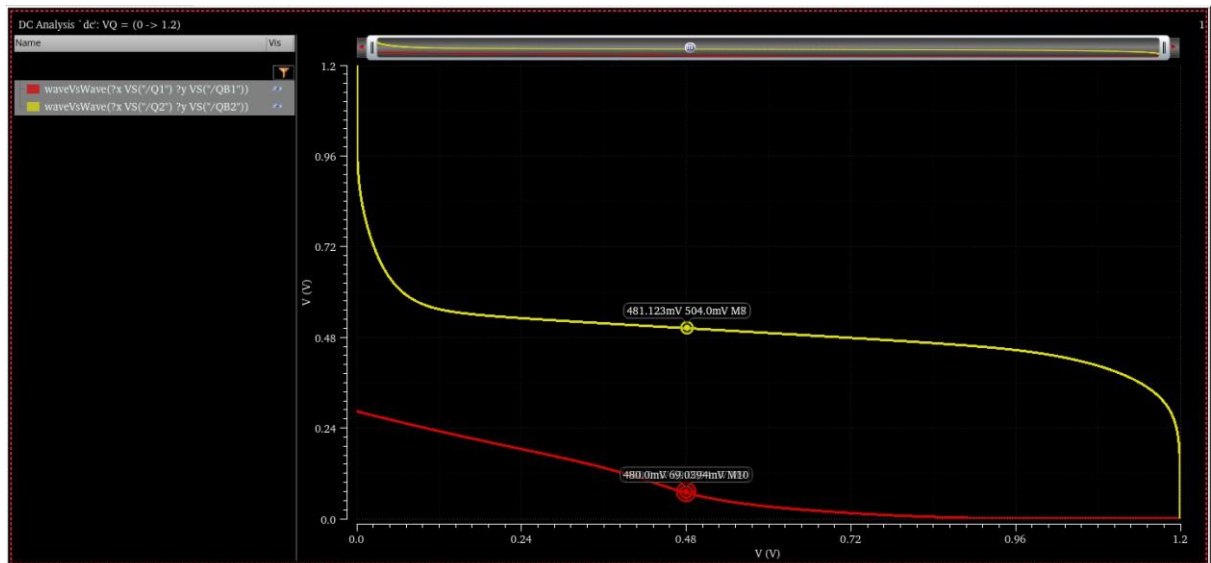


Waveforms:





Final Marker Values Used



Upper point on yellow curve: (0.4811, 0.5040)

Lower point on red curve: (0.4800, 0.0690)

Since the two points have nearly the same horizontal position, the write SNM was estimated from the vertical separation between the two curves:

$$SNM_{write} \approx 0.5040 - 0.0690$$

$$SNM_{write} \approx 0.4350 \text{ V}$$

Result

Therefore, the write static noise margin of the 6T SRAM cell is:

$$SNM_{write} \approx 0.435 \text{ V} = 435.0 \text{ mV}$$

The SNM values were estimated graphically from the plotted transfer characteristics using the final selected marker positions. The DC response of the 6T SRAM cell under write condition was obtained successfully. Because the bit lines were driven asymmetrically with $BL = 1.2$ V and $BLB = 0$ V, the two curves did not form the standard butterfly shape observed in hold and read conditions. Instead, the write SNM was determined from the largest square that could fit in the gap between the two curves. From the final selected marker positions, the write static noise margin of the SRAM cell was found to be approximately **435 mV**.

Final Summary

$$SNM_{write} \approx 435.0 \text{ mV}$$
[illegible]

Annexure

Netlists:

Q5) (a) Case 1: Hold Operation

VQ Sweepped:

```
// Point Netlist Generated on: Apr 20 23:06:37 2026
// Generated for: spectre
// Design Netlist Generated on: Apr 20 23:06:37 2026
// Design library name: HW02
// Design cell name: SRAM_Cell_TestBench
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27 VQ=0 VQB=0
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW02
// Cell name: sram6t_cell
// View name: schematic
subckt sram6t_cell BL BLB Q QB VDD VSS WL
M1 (QB Q VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M0 (Q QB VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M5 (QB WL BLB VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M4 (BL WL Q VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 ps=300n \
pd=300n ld=105n ls=105n m=1
M3 (QB Q VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M2 (Q QB VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
ends sram6t_cell
// End of subcircuit definition.

// Library name: HW02
// Cell name: SRAM_Cell_TestBench
// View name: schematic
I1 (net15 net17 Q2 QB2 net6 0 net9) sram6t_cell
```

```

I0 (net15 net9 Q1 QB1 net6 0 net17) sram6t_cell
V5 (QB2 0) vsource dc=VQB type=dc
V4 (Q1 0) vsource dc=VQ type=dc
V3 (net6 0) vsource dc=1.2 type=dc
V2 (net9 0) vsource dc=1.2 type=dc
V1 (net15 0) vsource dc=1.2 type=dc
V0 (net17 0) vsource dc=0 type=dc
simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
sensfile="./psf/sens.output" checklimitdest=psf
dc dc param=VQ start=0 stop=1.2 step=0.001 write="spectre.dc" \
oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
save Q1 QB1
saveOptions options save=allpub

```

VQB Sweepped:

```

// Point Netlist Generated on: Apr 20 23:10:44 2026
// Generated for: spectre
// Design Netlist Generated on: Apr 20 23:06:37 2026
// Design library name: HW02
// Design cell name: SRAM_Cell_TestBench
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27 VQ=0 VQB=0
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW02
// Cell name: sram6t_cell
// View name: schematic
subckt sram6t_cell BL BLB Q QB VDD VSS WL
M1 (QB Q VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1

```


b) Case 2: Read Operation

VQ Sweepped:

```
// Point Netlist Generated on: Apr 20 23:16:43 2026
// Generated for: spectre
// Design Netlist Generated on: Apr 20 23:16:43 2026
// Design library name: HW02
// Design cell name: SRAM_Cell_TestBench
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27 VQ=0 VQB=0
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW02
// Cell name: sram6t_cell
// View name: schematic
subckt sram6t_cell BL BLB Q QB VDD VSS WL
M1 (QB Q VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M0 (Q QB VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M5 (QB WL BLB VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M4 (BL WL Q VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 ps=300n \
pd=300n ld=105n ls=105n m=1
M3 (QB Q VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M2 (Q QB VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
ends sram6t_cell
// End of subcircuit definition.

// Library name: HW02
// Cell name: SRAM_Cell_TestBench
// View name: schematic
I1 (net15 net17 Q2 QB2 net6 0 net9) sram6t_cell
I0 (net15 net9 Q1 QB1 net6 0 net17) sram6t_cell
V5 (QB2 0) vsource dc=VQB type=dc
V4 (Q1 0) vsource dc=VQ type=dc
V3 (net6 0) vsource dc=1.2 type=dc
```

```

V2 (net9 0) vsource dc=1.2 type=dc
V1 (net15 0) vsource dc=1.2 type=dc
V0 (net17 0) vsource dc=1.2 type=dc
simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
sensfile="../psf/sens.output" checklimitdest=psf
dc dc param=VQ start=0 stop=1.2 step=0.001 write="spectre.dc" \
oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
save Q1 QB1
saveOptions options save=allpub

```

VQB Sweeoped:

```

// Point Netlist Generated on: Apr 20 23:19:05 2026
// Generated for: spectre
// Design Netlist Generated on: Apr 20 23:16:43 2026
// Design library name: HW02
// Design cell name: SRAM_Cell_TestBench
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27 VQ=0 VQB=0
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW02
// Cell name: sram6t_cell
// View name: schematic
subckt sram6t_cell BL BLB Q QB VDD VSS WL
M1 (QB Q VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M0 (Q QB VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M5 (QB WL BLB VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1

```

```

M4 (BL WL Q VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 ps=300n \
pd=300n ld=105n ls=105n m=1
M3 (QB Q VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M2 (Q QB VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
ends sram6t_cell
// End of subcircuit definition.

// Library name: HW02
// Cell name: SRAM_Cell_TestBench
// View name: schematic
I1 (net15 net17 Q2 QB2 net6 0 net9) sram6t_cell
I0 (net15 net9 Q1 QB1 net6 0 net17) sram6t_cell
V5 (QB2 0) vsource dc=VQB type=dc
V4 (Q1 0) vsource dc=VQ type=dc
V3 (net6 0) vsource dc=1.2 type=dc
V2 (net9 0) vsource dc=1.2 type=dc
V1 (net15 0) vsource dc=1.2 type=dc
V0 (net17 0) vsource dc=1.2 type=dc
simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
sensfile="../psf/sens.output" checklimitdest=psf
dc dc param=VQB start=0 stop=1.2 step=0.001 write="spectre.dc" \
oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
save Q2 QB2
saveOptions options save=allpub
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

```

c) Case 3: Write Operation

VQ Swept:

```

// Point Netlist Generated on: Apr 20 23:22:54 2026
// Generated for: spectre
// Design Netlist Generated on: Apr 20 23:22:53 2026

```

```
// Design library name: HW02
// Design cell name: SRAM_Cell_TestBench
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27 VQ=0 VQB=0
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"
```

```
// Library name: HW02
// Cell name: sram6t_cell
// View name: schematic
subckt sram6t_cell BL BLB Q QB VDD VSS WL
M1 (QB Q VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M0 (Q QB VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M5 (QB WL BLB VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M4 (BL WL Q VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 ps=300n \
pd=300n ld=105n ls=105n m=1
M3 (QB Q VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M2 (Q QB VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
ends sram6t_cell
// End of subcircuit definition.
```

```
// Library name: HW02
// Cell name: SRAM_Cell_TestBench
// View name: schematic
I1 (net15 net17 Q2 QB2 net6 0 net9) sram6t_cell
I0 (net15 net9 Q1 QB1 net6 0 net17) sram6t_cell
V5 (QB2 0) vsource dc=VQB type=dc
V4 (Q1 0) vsource dc=VQ type=dc
V3 (net6 0) vsource dc=1.2 type=dc
V2 (net9 0) vsource dc=0 type=dc
V1 (net15 0) vsource dc=1.2 type=dc
V0 (net17 0) vsource dc=1.2 type=dc
simulatorOptions options psfversion="1.4.0" reltol=1e-3 vabstol=1e-6 \
iabstol=1e-12 temp=27 tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 \
```

```
maxnotes=5 maxwarns=5 digits=5 cols=80 pivrel=1e-3 \
sensfile="./psf/sens.output" checklimitdest=psf
dc dc param=VQ start=0 stop=1.2 step=0.001 write="spectre.dc" \
oppoint=rawfile maxiters=150 maxsteps=10000 annotate=status
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
save Q1 QB1
saveOptions options save=allpub
```

VQB Sweepped:

```
// Point Netlist Generated on: Apr 20 23:25:44 2026
// Generated for: spectre
// Design Netlist Generated on: Apr 20 23:22:53 2026
// Design library name: HW02
// Design cell name: SRAM_Cell_TestBench
// Design view name: schematic
simulator lang=spectre
global 0
parameters temperature=27 VQ=0 VQB=0
include "/app11/FreePDK45/1.4/ncsu_basekit/models/hspice/hspice_nom.include"

// Library name: HW02
// Cell name: sram6t_cell
// View name: schematic
subckt sram6t_cell BL BLB Q QB VDD VSS WL
M1 (QB Q VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M0 (Q QB VDD VDD) PMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M5 (QB WL BLB VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M4 (BL WL Q VSS) NMOS_VTG w=200n l=50n as=9.45e-15 ad=9.45e-15 ps=300n \
pd=300n ld=105n ls=105n m=1
M3 (QB Q VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
ps=300n pd=300n ld=105n ls=105n m=1
M2 (Q QB VSS VSS) NMOS_VTG w=400n l=50n as=9.45e-15 ad=9.45e-15 \
```

